Cylindric Plane Partition

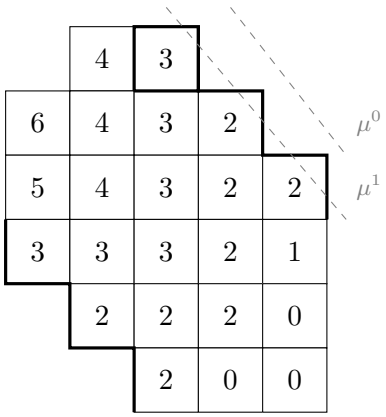
$$\mathfrak{c} = \big((3,2,2),\ (4,3,2,1),\ (4,3,2),\ (6,4,3,2),\ (5,3,2),\ (3,2,2)\big)$$

Profile $\pi = 10100$

Generated from thesis examples

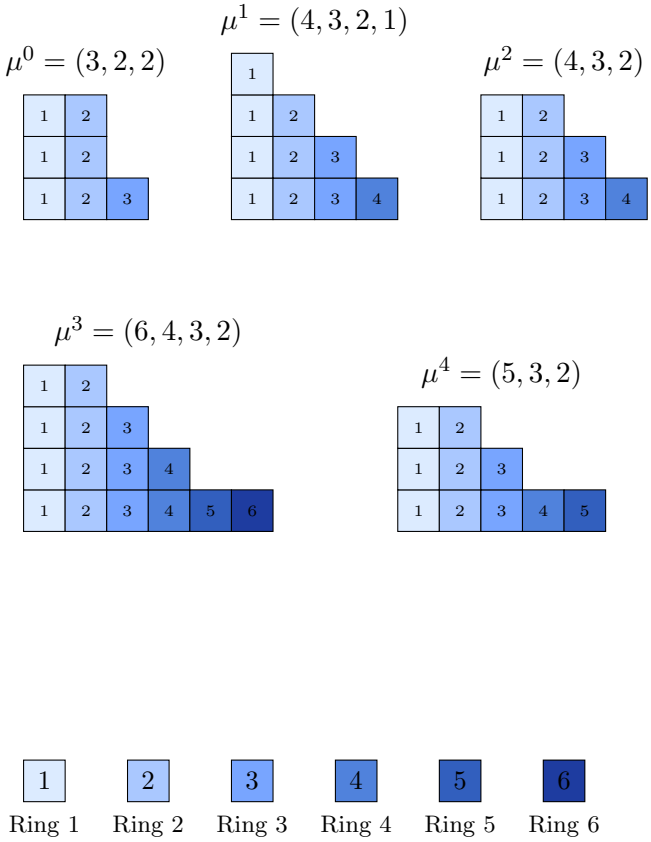
1 Labelled cylindric diagram (numbers on faces)

The fundamental domain of the cylindric diagram with profile $\pi = 10100$. Each entry gives the height of the cube stack at that position. The NW–SE diagonals (from right to left) read off $\mu^0 = (3,2,2)$, $\mu^1 = (4,3,2,1)$, $\mu^2 = (4,3,2)$, $\mu^3 = (6,4,3,2)$, $\mu^4 = (5,3,2)$.



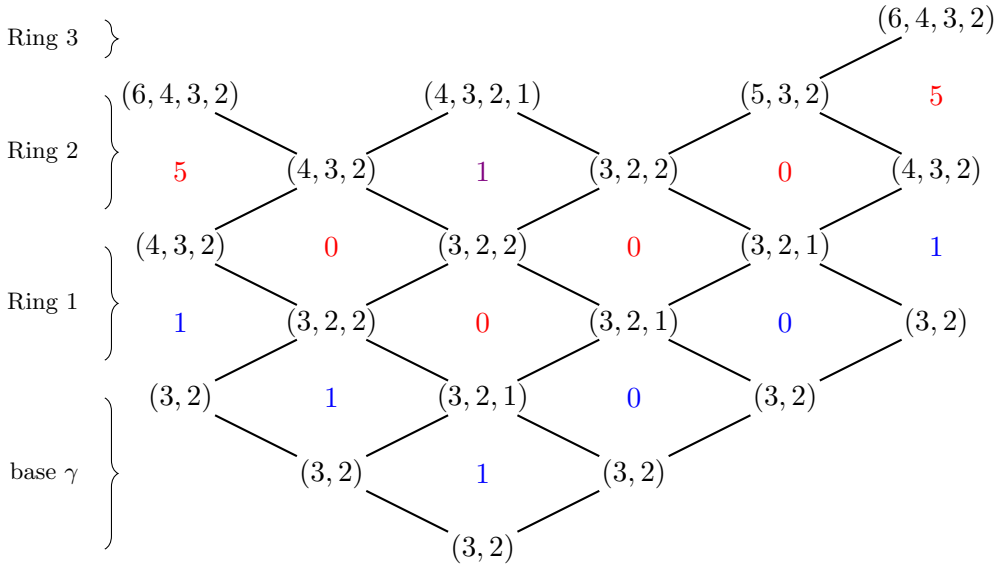
2 Ring colouring (PEEL)

Each cell in column j of a Young diagram belongs to **ring** j. The growth diagram has $\max(\mathfrak{c}) = 6$ concentric rings. Peeling ring k demotes all parts equal to k by one.



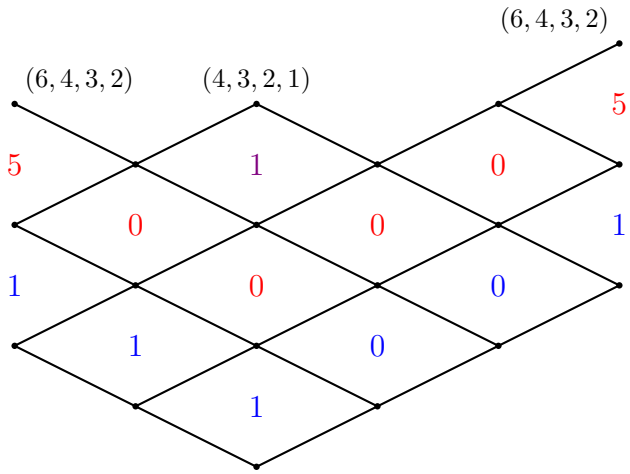
3 Growth diagram with ALCD

The cylindric growth diagram with profile $\pi = 00101$ (rotation of 10100). Vertex labels are partitions; face labels are the ALCD entries. Base partition $\gamma = (3,2)$.



4 The ALCD (face labels only)

Forgetting the vertex labels and retaining only the face labels gives the arbitrarily labelled cylindric diagram. Depth = 3, weight = $1 \times 1 + 5 \times 1 + 1 \times 4 + 1 \times 7 + 1 \times 9 = 26$.

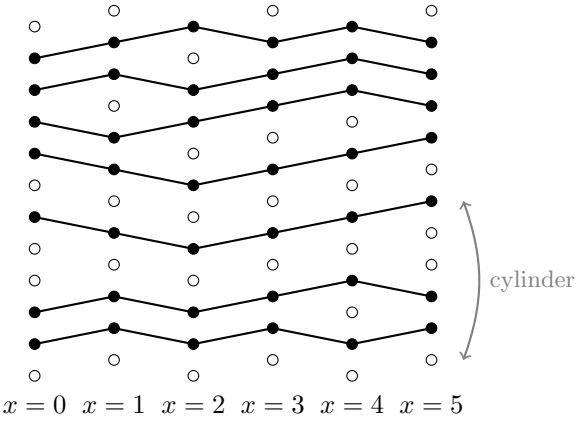


The ALCD in tableau form (on the cylindric diagram fundamental domain):

$$\begin{array}{ccccc} & 1 & 0 & & \\ 5 & 0 & 0 & 0 & \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ & 0 & 0 & 0 & 0 \\ & & 0 & 0 & 0 \end{array}$$

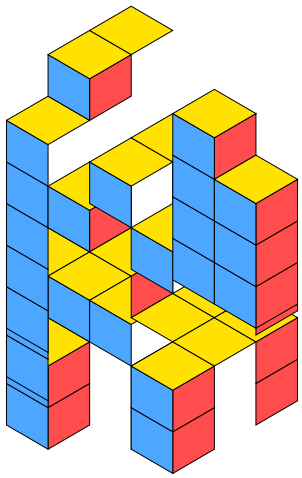
5 Non-intersecting lattice paths (page 81)

The CPP corresponds to a family of 7 non-intersecting lattice paths on the cylinder of period $T = 5$. Black vertices are occupied (path passes through); white vertices are unoccupied.



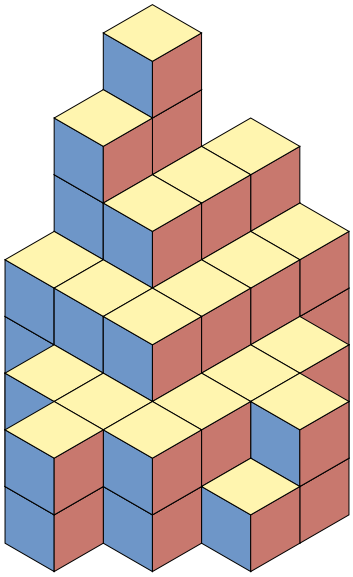
6 Rhombus tiling (page 81)

The 3D isometric view of the cylindric plane partition as stacked cubes. Yellow = top faces, red = right faces, blue = front faces.



7 3D isometric view (cubes stacked on squares)

Each square of the labelled cylindric diagram with label k is drawn as a stack of k unit cubes, viewed from an isometric (axonometric) angle. The three visible faces of each cube are coloured: yellow (top), blue (left), red (right). Cubes are drawn back-to-front (painter's algorithm) sorted by depth $c + r$, so that nearer cubes correctly occlude farther ones.



8 Cylindric repetition (two periods)

The profile $\pi = 10100$ has $n = 2$ ones and $m = 3$ zeros, so each period the staircase shifts by $(+2, -3)$ in (column, row). The fundamental domain (Section 7) is shown together with one additional period to the upper right, illustrating the cylindric structure.

